

Article

North Florida Stakeholder Perspectives: Gaps, Barriers, and Facilitators in the Transportation System

Mary Jeghers , Sandra Winter  and Sherrilene Classen

Department of Occupational Therapy, University of Florida, Gainesville, FL 32611, USA

* Correspondence: maryjeghers@gmail.com

Abstract: Florida's population is projected to grow by 8.8 million residents over the next 25 years. This increase places demands on the transportation system, particularly for mobility-vulnerable populations, potentially impacting equitable transportation options and access for all users. Developing transit solutions for mobility-vulnerable populations may enhance economic activity, health, and well-being. Inquiry is needed to explore transportation stakeholders' strategies and perspectives on the challenges and opportunities of the existing transportation system. Therefore, this study examined stakeholders' views on strategies to address user needs, related challenges, and opportunities while identifying gaps, strengths, weaknesses, and threats relevant to addressing transportation among mobility-vulnerable populations. The team conducted 13 semi-structured interviews with city planners, transportation employees, industry stakeholders, and state representatives. Findings indicate gaps in the transportation system, particularly available funding and limited common goals among stakeholders. Participants emphasized the need for enhanced educational resources and collaboration with community members. They identified strengths like a willingness to pilot innovative transit technologies, weaknesses such as unreliable options, opportunities for innovation, and threats, including COVID-19's impact on transportation use. Understanding transportation stakeholders' shared challenges and opportunities is crucial for identifying transit needs and developing strategies to reduce disparities for mobility-vulnerable populations.

Keywords: community mobility; transportation planning; transportation stakeholders; mobility-vulnerable populations



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1. Introduction

Florida's population in 2023 was 22.6 million, with projections estimating it could reach 31.4 million by 2050—an annual increase of approximately 326,000 residents [1]. An influx in population growth places increased demands on the transportation system, potentially impacting equitable transportation options and access for all users. Access to personal vehicles may not be feasible for all residents. For example, mobility-vulnerable populations—whose incomes are below the federal poverty threshold, individuals with disabilities, and older adults—have limited transportation options to meet their basic needs, impacting productivity and quality of life [2]. Transportation stakeholders have started implementing innovative alternatives to address transportation disparities, such as rideshare services [3]. Addressing transit challenges and opportunities for mobility-vulnerable populations is an urgent need for transportation stakeholders [4]. Creating flexible transportation solutions, especially for mobility-vulnerable populations, may support economic activity, health, and well-being.

Therefore, exploring transportation stakeholders' strategies to meet users' needs—and their perspectives on transportation-related challenges and opportunities—requires scientific inquiry. The purpose of this study was to conduct semi-structured interviews with city planners, transportation agency employees, industry stakeholders, and representatives from the Florida Department of Transportation (FDOT) to gather their perspectives. The

findings are structured around three key areas: (1) identifying current gaps; (2) stakeholders are implementing strategies to meet the diverse needs of transportation users; and (3) assessing the strengths and weaknesses and examining the opportunities and threats essential to addressing transportation disparities among mobility-vulnerable populations.

1.1. Review of the Literature

1.1.1. Transportation in Florida

Florida has more than 22.6 million residents and approximately 135 million visitors annually [5]. Based on the 2018 American Community Survey statistics among Florida residents, 79.1% drove personal vehicles, 9.4% carpooled, 1.7% used public transit, 1.4% walked, 0.6% biked, 0.6% used an alternative form of transit, and 6.2% worked from home and did not have to commute to work [6]. Although commuting to work only constitutes part of a community's travel demands, insight into how residents access their community provides valuable information—i.e., emphasizing the car-dependent society in Florida for the daily use of transportation.

1.1.2. Transportation Challenges and Disparities in Florida

One of the largest mobility-vulnerable populations in Florida is older adults (65+ years), who are expected to grow by 60% by 2045 [6]. Older adults may prefer to use a personal vehicle compared to other forms of transit due to its convenience and flexibility [7]. Florida has 3,000,000 drivers, 65 years or older, and 90% access their community using a personal vehicle [8]. As older adults experience age-related declines and retire from driving, the demand for alternative transportation modes increases. A survey of 3840 Florida residents aged 65 and older found that 42% plan to rely on family or friends for rides, 32% will walk occasionally, and 40% intend to use rideshares like Uber or Lyft once they retire from driving [9]. As Florida's older adult population grows, accessible and flexible transportation alternatives will be needed to maintain independence and quality of life.

Other mobility-vulnerable populations, such as people with disabilities and individuals living below the poverty line, rely on public transportation or alternative transit, like paratransit or bicycling. Of the 25 million people with disabilities, about 13 million Americans (52.7%), ages 18 to 64, report having a travel-limiting disability [10]. In addition, over half of the people aged 18 to 64 with disabilities live in households with income levels under USD 25,000. Therefore, many individuals with a travel-limiting disability (3.6 million Americans) are homebound or lack access to travel accommodations [11]. Considering mobility-vulnerable populations is crucial, as many lack access to well-connected transit systems or the social and financial support needed for transportation. This study investigated strategies used by transportation stakeholders (from various entities—city, state, and federal transportation organizations, advocacy groups, and researchers) to meet users' needs. Additionally, stakeholders' perspectives on the transportation system's gaps, strengths, weaknesses, opportunities, and threats were explored.

Transportation stakeholders identified smart city mobility options as a potential solution to help mitigate transportation disparities for mobility-vulnerable populations [12]. Smart mobility includes transportation modes, such as ride-hailing and autonomous ridesharing services, designed to improve the efficiency, accessibility, and sustainability of transportation systems [13]. While much of the literature on smart cities focuses on technological advancements, less attention is given to how these innovations impact mobility-vulnerable populations [14]. Gainesville, Florida, is part of the Smart Cities Collaborative, which seeks to address transportation inequities through innovative technologies like ridesharing. A study in Gainesville, Florida [15] explored transportation challenges and opportunities among residents of low- and high-socioeconomic status (SES) communities to inform recommendations for implementing new transit technologies. Focus group findings indicated that participants from the low-SES community found challenges with bus accessibility, cleanliness, and user-friendliness. In contrast, participants in the high-SES communities were more concerned with traffic congestion and parking. Participants in

both communities expressed safety concerns (e.g., crime) and limited bus service times and routes but found bus fares affordable. Addressing transit disparities requires collaboration and innovative thinking from various transportation stakeholders, who play a crucial role in shaping the future of mobility solutions.

1.1.3. Transportation Stakeholders

The team explored the perspectives of various transportation stakeholders to capture a depth of knowledge regarding transportation challenges and opportunities. The transportation stakeholders included city and state entities (e.g., FDOT), the private sector (e.g., representatives from organizations operating various transit modes, including buses, paratransit, and autonomous vehicles), advocacy groups (e.g., AARP), and representatives of mobility-vulnerable populations, specifically older adults and individuals with disabilities.

Effective communication between community members, transportation personnel, and other stakeholders is important for understanding and addressing transportation needs and preferences [16]. Each stakeholder has a distinct role in shaping transportation use and investments [17]. Community residents may value access and ease of transportation, whereas administrators and city officials may prioritize optimizing safe transportation in a city. In Florida, initiatives like the Safe Mobility for Life Coalition demonstrate the potential for stakeholders to collaborate on improving transportation for mobility-vulnerable populations [18]. This coalition includes members from statewide organizations, health-care professionals, and academic institutions working together to address transportation challenges. By incorporating diverse stakeholder perspectives, this model demonstrates how collaborative efforts may lead to effective, equitable, and sustainable improvements in transportation systems, as outlined in this study.

1.2. Rationale and Significance

Disparities in transportation exist on a national, statewide, and local level. Understanding transportation user needs and provider perspectives is crucial for implementing the best strategies to mitigate transportation challenges. Prior research on Gainesville residents highlighted transportation challenges, such as limited options and service times, and opportunities, like a fixed-route bus and reliance on personal vehicle rides [15]. Insights from the participants in the Jeghers et al. [15] study contribute to understanding the users' needs. Nevertheless, a similar approach (i.e., in-depth exploration) is necessary to solicit the perspectives of city planners, transportation agency employees, industry stakeholders, and representatives of the FDOT. In doing so, the research team identified current transportation gaps, strengths, weaknesses, opportunities, and threats, which are critical for understanding how to address transportation disparities among mobility-vulnerable populations.

1.3. Purpose

The purpose of this study was to conduct semi-structured interviews with transportation stakeholders to understand the multiple factors involved in equitable transportation options for the residents of Florida. The research team sought to answer the following questions:

1. From the stakeholders' perspectives, what gaps exist in Florida's current transportation system?
2. What strategies are being used by stakeholders in Florida to meet the needs of transportation users?
3. From stakeholders' perspectives, what are the current transportation system's strengths, weaknesses, opportunities, and threats?

The team used Barrella et al.'s [19] definition of strengths, weaknesses, opportunities, and threats (SWOT) to provide established guidelines when discussing the transportation system challenges and opportunities. Strengths and weaknesses were defined as internal environmental characteristics, whereas opportunities and threats referred to external characteristics of the transportation system.

2. Methods

2.1. Ethics

The study received approval from the University of Florida's Institutional Review Board (UF IRB-02).

2.2. Study Design

Qualitative semi-structured interviews to gain in-depth insights from Florida's statewide transportation stakeholders.

2.3. Participants

The participants (N = 13) included regional transportation services employees, city planners and employees, state employees, industry employees, transportation advocates, and end users. Employees of organizations that serve people with disabilities, aging adults, and advocates of the Veteran population were also recruited. See Table 1 (located in Results) for a list of participants and their roles.

2.3.1. Recruitment

The team utilized purposive sampling by choosing a specific sample of participants based on pre-determined criteria [20]. Additionally, snowballing techniques were utilized where existing participants recruited acquaintances to participate [21]. Through leveraging existing networks and researching online public records, the team developed a list of potential participants to recruit. Participants were contacted via email or phone to inquire about participation and complete the eligibility screening. A pre-determined screening script was utilized, including the study's purpose, the procedure of the semi-structured interviews, an overview of interview questions, inclusion/exclusion criteria screening questions, expectations of time commitments, and follow-up plans after the interview.

2.3.2. Eligibility Criteria

Individuals eligible to participate met the following criteria: they were associated with transportation services for the City of Gainesville or the state of Florida that serve mobility-vulnerable populations; they had access to reliable and secure internet, including Zoom videoconferencing; and they could read, speak, and write English. Due to COVID-19 and social distancing requirements, participants needed access to Zoom videoconferencing. Individuals who could not adequately speak to the transportation system for Gainesville or Florida were excluded.

2.3.3. Sample Size

The team aimed to recruit 15 transportation stakeholders to increase the likelihood of data saturation where no new data emerged [22].

2.4. Procedure

2.4.1. Development of a Semi-Structured Interview Guide

The team developed an interview guide to investigate Gainesville's transportation challenges and opportunities identified in the Jeghers et al. [15] study and explore transportation experts' perspectives on strengths, weaknesses, opportunities, and threats. See Supplementary Materials for a copy of the Semi-Structured Interview Guide.

2.4.2. Setting

The semi-structured interviews were conducted during the COVID-19 pandemic (2021). Thus, all interviews were conducted in a virtual format via Zoom to adhere to the precautions for COVID-19 social distancing [23].

2.4.3. Semi-Structured Interviews

Participants completed an electronic informed consent and a demographic questionnaire using Qualtrics. The team utilized the university’s Zoom subscription and provided training on using Zoom features before the interview. Utilizing the semi-structured interview guide, the team welcomed and oriented participants (5 min), discussed interview questions and prompts (50 min), closed the interview with an explanation of the following steps, and thanked participants for their time (5 min).

2.5. Data Collection

Each semi-structured interview was audio-recorded using Zoom. Recordings, demographic questionnaires, and informed consent forms were saved to a password-protected file on the University’s server. To maintain anonymity, all participants were assigned a unique identifier.

2.6. Data Management

To improve rigor and validity, a third-party service was used to complete the first round of transcription for each interview [24]. Then, the team reviewed each transcript to ensure completeness and accuracy. Transcripts were transferred to NVivo Qualitative Data Analysis software (Version 12) [25] for data analysis.

2.7. Data Analysis

The demographic questionnaire data were input in SPSS 25.0 [26]. Demographic data were analyzed using descriptive statistics, specifically frequency distributions. The team conducted a content analysis on each transcript, including identifying, coding, and counting participants’ perceptions of the transportation system’s SWOT [9,27].

3. Results

3.1. Semi-Structured Interview Participant Demographics

Thirteen transportation stakeholders participated in the semi-structured interviews. Participant demographics, including their stakeholder position and experience(s) within the transportation sector, are detailed in Tables 1 and 2.

Table 1. Transportation stakeholders from sectors recruited for the semi-structured interviews.

Transportation Sector	Number of Stakeholders Recruited
Florida State Department of Transportation (FDOT)	2
Research-related	
Engineering-related	
Gainesville City and Regional Transportation Administration	2
Director and managerial roles	
City planner	1
Advanced technologies—Private sector	
Advocacy groups	
Gainesville Citizens for Alternative Transportation	2
Regional Mobility Group	
Specialty population—Aging	2
AARP Florida Age-Friendly Network	
Pepper Institute on Aging and Public Policy	
Specialty population—Veterans	1
Malcolm Randall Department of Veterans Affairs Medical Center	
Specialty population—People with disabilities	3
Florida Commission for the Transportation Disadvantaged	
ADA Transit (2)	

Note. Ten of the 13 eligible participants completed the demographic questionnaire, and three declined.

Table 2. Demographics of the transportation stakeholders (N = 10).

Demographic Characteristic	Number of Participant Responses (% of Participants)
Geographic areas the stakeholder group serves *	
Urban	8 (80)
Suburban	7 (70)
Rural	6 (66)
Other—Statewide	1 (10)
Other—Undeclared	1 (10)
Length of time participant was employed in the transportation sector	
Less than a year	1 (10)
One to five years	2 (20)
Six to ten years	0 (0)
Over ten years	7 (70)

Note. * Some individuals answered more than one option per demographic characteristic, therefore the number of participant responses add up to more than 100%.

3.2. *Semi-Structured Interview Discussion Findings*

Participant responses from the semi-structured interviews are summarized below.

3.2.1. *Gaps in the Transportation System*

The findings indicate several key themes regarding the transportation system’s current gaps and challenges, including (1) funding constraints, (2) priorities of state and city officials, (3) addressing the needs of mobility-vulnerable populations, and (4) transportation connectivity issues. Each theme underscores a distinct aspect of the transportation challenges residents or stakeholders face. To further support the findings, participant responses from the semi-structured interviews are summarized below, and direct quotes from the stakeholders are provided in Table 3.

Table 3. Themes and Quotes Related to Gaps in the Transportation System.

Transportation System Gap	Example Provided by Participant(s)	Supporting Quote(s)
Funding	- Limited ability to relocate funds to a higher priority/need once grant funds were allocated - Impacts of uncertain future funding on long-term planning	“The challenge is not having any control of the funding sources. . .and not knowing what kind of funding we will have the following year”. (P2) “It is nice to have five to 10-year long-term plans, but you do not have dedicated funding to provide those services to execute those plans [on a long-term basis]. Then that leaves you with service gaps, and you see complaints [from users]”. (P4) “They [city and county commissions] are looking at what is the minimum that can be done to make it [transportation operations and infrastructure] as safe as possible without having to spend real money”. (P3)

Table 3. Cont.

Transportation System Gap	Example Provided by Participant(s)	Supporting Quote(s)
Priorities of State and City Officials	- Importance of collaboration among federal, state, and city entities to achieve shared goals - Impacts of misaligned goals of transit users and city and state officials (e.g., prioritization of student ridership frequency and funding compared to other populations in Gainesville, including those in less populated areas or higher rates of minorities)	“Politicians come and go. They have priorities, and they do not always match [the priorities of multiple stakeholders] because the East Side [area in Gainesville, FL with a higher percentage of minority population] is still hurting”. (P13)
Addressing the Needs of Mobility-Vulnerable Populations	- Residents below the federal poverty threshold faced limited public transportation routes and longer service times - Low ridership during the COVID-19 pandemic impacted both transit riders and transportation employees - Limited adaptability to accommodate older adults or people with disabilities (e.g., use of mobility aides)	“During COVID-19, bus operators for public buses and alternative services for the Veteran population were furloughed or released from their duties to accommodate reduced ridership demands”. (P10) “Many cities have public transportation, but for older adults with a cane, [they] may have balance issues. They have a hard time getting up the steps [to enter and exit the bus]...make it more accessible by lowering the bus so that it [the bus] meets the sidewalk and is easy to get on and off”. (P9)
Transportation Connectivity	- Lack of connectivity within the transportation system - Limited wheelchair-accessible sidewalks and increased safety concerns with inadequate spaces for walking	“Road features, such as cul-de-sacs in residential areas, increased the time necessary for persons to access the main roadways or bus stops. This, in turn, made transportation less accessible for users in those areas”. (P3) “There is no point in having an accessible bus if a person with a wheelchair can’t get to the bus stop or a person who is blind has no audible information to navigate the bus system. To indicate that the paratransit is an acceptable alternative to mainstream transportation, [when it] isn’t accessible, is an affront to those with disabilities who are otherwise independent in their activities”. (P6)

Funding

Ten of the thirteen participants stated that funding was one of the most significant gaps in transportation, specifically grant funding and the costs of operation, maintenance, and transportation infrastructure improvements. Participants discussed that grant funding was provided to meet specific needs, such as improving roadways, traffic congestion, or increasing transit services. However, once the funds were allocated, the ability to relocate the funds was often restricted, even if a higher priority need was identified (e.g., locations with deficient bus service). Participants also explained the challenges of long-term planning due to uncertainties around funding. Multiple participants articulated that funding directly impacted what a city could offer regarding transportation services and infrastructure—and was a key factor leading to transportation gaps.

Priorities of State and City Officials

Four of the thirteen participants mentioned the importance of state and city officials coordinating efforts toward common goals. Several participants provided examples of how federal, state, and city entities work together to achieve shared goals, such as Target

Zero—an initiative to eliminate traffic deaths, reduce injuries, and increase safety and equitable mobility for all [18]. In contrast, other participants discussed the impacts of transportation disparities (e.g., limited bus routes in areas with a higher proportion of minorities) due to misaligned goals of transit users and city and state officials. Given the presence of the University in Gainesville, participants explained city officials prioritize student ridership frequency and funding compared to other populations in Gainesville. Furthermore, focusing on university student ridership may reduce services in less-populated or minority-dense areas, creating inequities.

Addressing the Needs of Mobility-Vulnerable Populations

Five of the thirteen participants described transportation gaps by providing specific challenges that mobility-vulnerable populations face. Residents below the federal poverty threshold encountered limited public transportation routes and longer service times. The COVID-19 pandemic worsened these gaps by reducing available services. Therefore, users relied on family or friends for rides or found options that may be more expensive and less feasible, such as ridesharing. Adaptability concerns for older adults and individuals with disabilities, including bus design and inadequate sidewalk infrastructure, were also discussed, indicating the need for more inclusive transportation planning.

Transportation Connectivity

Among the gaps, four of the thirteen participants discussed the lack of connectivity within the transportation system. Participants centered the discussion around the lack of transportation infrastructure, specifically the lack of “complete streets”—planned and operated streets that enable safe, convenient, and comfortable travel for users of all ages and abilities [16]. Participants discussed infrastructure like cul-de-sacs and inaccessible sidewalks, which limit mobility for people using wheelchairs and people with visual impairments. Initiatives like Complete Street may address connectivity limitations and improve safety for sidewalk and roadway users. However, the needs of all users, including mobility-vulnerable populations, should be considered while comprehensively planning transportation systems.

In conclusion, transportation gaps discussed among participants include the current lack of funding and the establishment of common goals among state and city officials and community members to enhance transportation. Without these efforts, stakeholders may face challenges in developing and sustaining an effective transportation system, potentially leading to limited opportunities and connectivity for mobility-vulnerable populations. To further understand the gaps and barriers that users experience, transportation stakeholders often evaluate the system and create strategies to meet community needs, next discussed.

3.2.2. Use of Strategies to Meet the Needs of Users

Participants discussed strategies to address challenges and improve the transportation system, including aligning goals across stakeholders, educating the public about transportation options, and providing user hands-on training. Table 4 details the main themes discussed by participants, along with supporting quotes that illustrate the importance of these initiatives in creating a transportation system that serves all populations, including mobility-vulnerable people.

Participants emphasized the importance of aligning goals across all levels of stakeholders and implementing comprehensive educational strategies. Strategies include public awareness campaigns, resources, and hands-on training programs to educate transit users, especially mobility-vulnerable populations. Discussion points also centered around challenges from lacking community awareness of available resources, underscoring the need for ongoing education and outreach. Findings also discuss two critical educational resources in Florida: (1) the FDOT Safety Agency Resource (<https://www.fdot.gov/agencyresources/resources>, accessed on 13 July 2022), which offers information on topics such as bicycle safety and impaired driving; and (2) the Safe Mobility for

Life Coalition (<https://www.safemobilityfl.com/>, access on 11 July 2022), which provides resources for older adults to enhance safe mobility. Federal initiatives aim to educate stakeholders on providing accessible and equitable transportation. However, participants identified gaps, such as resources often focusing on wheelchair accessibility without adequately addressing other disabilities (e.g., vision deficits).

Table 4. Strategies to Meet the Needs of the Users.

Theme	Identified Strategy	Supporting Quote
Goal alignment	Alignment of goals among all stakeholders is key for community-focused transportation improvements	“We [city transit employees] are out in the community daily. . . trying to find creative ways to get feedback.” (P2)
Community education and resources	Educating the community about transit options and safety measures (e.g., helmet use)	“It’s not as though most people in Florida know about what we [coalition members] do.” (P8)
Educational resources	Resources provided by state entities (e.g., Safe Mobility for Life Coalition) may provide vital safety information	“The webinars focus primarily on individuals with wheelchairs. . . they do not go into individuals who are deaf or blind.” (P6)
Community engagement	Stakeholders use various feedback methods, including surveys, community events, and phone or tablet applications	“Using website feedback forms or a smartphone app. . . has successfully worked to obtain feedback.” (P4)
Hands-on training	Programs like travel training and partnerships with other transit stakeholders, may help users, such as people with disabilities, gain skills to use transportation independently	“AARP partners with Safe Mobility for Life. . . where we [stakeholders] try to help people feel more comfortable taking the bus.” (P9)

To engage the community and gather feedback, stakeholders employed diverse methods, such as surveys and real-time feedback via smartphone apps, and attended community events. These efforts allowed transportation organizations to adapt to the evolving needs of users and adjust strategies to mitigate challenges. Hands-on training, such as travel training programs facilitated by partnerships with organizations like AARP, may promote greater independence and confidence among transit users. Such initiatives are essential for fostering a transportation system that meets the diverse needs of all community members, particularly mobility-vulnerable populations. One of the strategies used by transportation stakeholders was to develop resources and educate users on the existence and use of different transit options. Participants discussed the importance of educating the community about transportation options and transit safety (e.g., helmet use), in addition to soliciting feedback to understand the challenges and opportunities of the transportation system.

Participants discussed ways their stakeholder organizations gathered information from communities, such as interviews, discussions, surveys, and ridership data. Developing strategies to help mitigate user challenges was commonly discussed, specifically related to educational resources and hands-on training tailored to various populations and needs.

3.2.3. Strengths, Weaknesses, Opportunities, and Threats of the Transportation System

To further understand the factors contributing to developing and implementing equitable transportation systems, the participants discussed their perspectives on the strengths, weaknesses, opportunities, and threats. Table 5 provides an overview of the factors discussed and the number of participant references.

Strengths of the Transportation System

Among the transportation system strengths (15/101 counts), participants introduced innovative transportation technologies (i.e., autonomous rideshare services), transportation availability in various geographical areas, and transportation stakeholders’ expertise and goals. Participants mentioned that Gainesville city officials and transit stakeholders were willing to try innovative technologies that may promote a more equitable transportation system for mobility-vulnerable populations.

Table 5. Strengths, weaknesses, opportunities, and threats discussed among the participants.

SWOT Analysis	Factors Mentioned by Participants	Number of References Made by Participants	Supporting Quote(s)
Strengths	Implementing innovative transportation technologies	5	“There is a lot of innovation happening, and there are a lot of new modes [of transportation] that I think will help provide older adults with other opportunities so they can turn in that key when they’re ready to do so [retire from driving].” (P9) “One of the biggest strengths is the staff that’s here [in Gainesville]. . . [the traffic operator] is able to work with the University and researchers for all these different emerging technologies. . .” (P2)
	Transportation options for geographic areas (e.g., rural, suburban, urban)	4	
	The current transit system builds a strong foundation	3	
	Knowledgeable transportation stakeholders and city planners	2	
Weaknesses	City officials support	1	“I have a general sense that the public transportation options outside of private transit. . . there aren’t a lot of options. I think the bus route is it unless you. . . have your own private transportation or you engage in some type of ridesharing.” (P10) “We don’t have so many alternatives to driving. So, the people who aren’t comfortable driving, can’t afford to drive. . . don’t have many options in terms of buses or well-networked bus routes.” (P8)
	Lack of services to vulnerable populations	15	
	Unable to meet the needs of transit users	5	
	Transit providers violating American Disabilities Act (ADA) regulations	4	
	Unwillingness of users to change behavior	4	
	Lack of knowledge of transportation options	4	
	Dissatisfaction among transit operators	4	
Opportunities	Safety	2	“Florida is unique that we are one of the few states with a dedicated trust fund for this service [transportation], and the revenues are growing. The population is growing. Florida’s in a good position to continue to support that.” (P11) “There is a partnership with Walgreens, where they sell monthly, three-day, or five-day passes. . . They [mobility-vulnerable individuals] can pay cash [instead of relying on bank accounts or credit cards].” (P12) “We’ve tried to reduce costs, so opportunities like Uber or Lyft complement our services so we can reduce and use resources somewhere else.” (P4)
	Advancements with technology	1	
	Common objectives and goals among state and federal entities	10	
	Coordination and collaboration with multiple entities	10	
	Educating the community and users on transit options	3	
	Data collection and access to data to impact decision making	3	
	Funding	3	
	Support from community members to advocate for transportation	2	
	Development of resources to help users (e.g., scheduling assist)	4	
		1	
Threats		1	“How many people are riding around without the helmets on the [electric] scooters. . . How are they [electric scooters] going to hold up in crashes and. . . those kinds of safety issues. . .” (P8) “I still think we are developing a lot of different modes. . . we still have a lot of places that there are such gaps in modes of transportation, particularly for older adults.” (P9) “COVID has impacted us a lot. . . We saw a reduction of 70% in ridership, so we have to adjust the service accordingly.” (P4)
	Unknown impacts of transportation advancements	9	
	Impacts of COVID-19	7	
	Large corporations buying out transit operators	3	
	Cyber-security	1	

Weaknesses of the Transportation System

A primary weakness identified by participants, one of the most prominent points of discussion (32/101 counts), was the lack of transportation options for mobility vulnerable populations. Limited bus routes and inadequate pedestrian and bike infrastructure were weaknesses, particularly in rural areas. Participants discussed the challenges faced by individuals who cannot drive due to financial limitations, age, or disability and the lack of alternative transit options to meet their needs. Therefore, considering the weaknesses in Gainesville's transportation system, transportation stakeholders may focus on solutions to increase options (e.g., paratransit or use of public bus for individuals that are blind) and services (e.g., paratransit in rural areas) among mobility-vulnerable populations. Addressing the identified weaknesses may help create a more inclusive transportation system to better serve all residents, especially those most reliant on public or non-private transportation.

Opportunities of the Transportation System

Participants identified several opportunities, such as transportation advancements and the collaboration of multiple transportation entities to implement innovative transportation changes (e.g., autonomous shared mobility services; 34/101 counts). Notable opportunities include Florida's dedicated trust fund for transportation, which provides a foundation for growth and system-wide improvements. Additionally, collaborations with private organizations, such as Walgreens, to sell transit passes and integrate rideshare services like Uber and Lyft were seen as ways to address gaps in public transportation services. The opportunities for expansion and improvement of the transportation system within Gainesville include Florida's economic position and partnering with organizations and companies to offset limitations to mobility options.

Threats of the Transportation System

The discussion of threats included the unknown impacts of implementing new transportation technologies, such as automated vehicles, to mitigate disparities (20/101 counts). Participants identified concerns about the safety and regulatory policies for new transportation modes and how novel modes will impact mobility-vulnerable populations. Specifically, participants discussed the lack of protective measures (e.g., helmets) for riders and the challenge of ensuring these modes are accessible to older adults or those with disabilities. Additionally, the COVID-19 pandemic was cited as a significant threat, causing a reduction in ridership, particularly among university students, which necessitated adjustments to the transit system. These threats highlight the need to consider safety, accessibility, and public health factors as new technologies and challenges are introduced.

The participants had overlapping opinions regarding the transportation system's strengths, weaknesses, opportunities, and threats. Additionally, participants suggested the need to ensure that federal, state, and city stakeholders work together to develop plans to meet transportation users' needs.

4. Discussion

This study used 13 semi-structured interviews with transportation stakeholders to explore the gaps in the transportation system, strategies used to meet transportation users' needs, and the transportation system's strengths, weaknesses, opportunities, and threats.

4.1. Gaps in the Transportation System

Nearly all participants stated that funding is a critical gap in the Florida transportation system. Findings indicate that funding opportunities are often allocated based on federal and state priorities and are not long-term, creating challenges for planning and implementing transportation changes [28]. Participants discussed that stakeholders know about transportation inequities, specifically for mobility-vulnerable populations, but feel there are constraints to making changes to alleviate transportation barriers. Similarly, data from the Cantilina et al. [29] study indicate that funding is one of the biggest barriers

to transportation equity, more so for the public than private agencies. Funding barriers may exist due to misaligned objectives of public and private transportation entities. In the past, federal transportation funds were often allocated for projects on highways and roadways [30], which did not consider the funding needed to enhance the unique needs of mobility-vulnerable populations (e.g., paratransit service, bus stop maintenance). Obtaining information from various stakeholders and developing strategies may support the development of common objectives [31]. Findings illustrate that the stakeholders were often aligned in their opinions of the transportation system. However, there is still a need to engage stakeholders to establish shared goals and visions for community transportation.

4.2. Use of Strategies to Meet the Needs of Users

The participants shared strategies to assist with meeting users' needs with education-based approaches, such as developing resources, travel training, and partnering with large corporations to hold webinars. Previously, researchers have found success using less traditional approaches, such as talking to transit users at frequented community locations [32,33]. Hosting events at convenient times and locations for community members boosted public involvement and may be more accessible for individuals with limited internet access. Additionally, Evans-Cowley and Griffin [34] found benefits in using social media platforms (e.g., Facebook) to engage with the community and understand users' perceptions of transportation challenges and opportunities. Transportation stakeholders may leverage social media platforms to help educate community residents about transportation system initiatives [35]. The study findings indicated the importance of bringing awareness of public transportation options in the community, and they may help individuals planning on retiring from driving or mobility vulnerable populations. Based on the interview data, transportation stakeholders know of the gaps and transit needs for mobility-vulnerable populations. With this knowledge, transportation stakeholders may focus on creating meaning for the identified gaps, transit needs, and subsequent solutions to foster equitable transportation systems.

4.3. Strengths, Weaknesses, Opportunities, and Threats of the Transportation System

4.3.1. Strengths

The strengths discussed among the semi-structured interview participants included innovative transportation technologies (i.e., autonomous rideshare), transportation availability in various geographical areas, and transportation stakeholders' expertise and goals. These findings are particularly relevant to the local transportation system in Gainesville, Florida. The City of Gainesville participates in the Smart Cities Collaborative [36] and has implemented new modes of transportation (i.e., autonomous rideshare) to improve mobility in the community and serve the needs of mobility-vulnerable populations. In 2014, Florida adopted the Complete Streets Policy statewide to promote safety, quality of life, and modal transportation choices [16]. These initiatives at the city and state levels indicate the desire of stakeholders to enhance the transportation system and work with other stakeholders, such as funders, advocates, and transit users, to develop equitable systems for all users.

4.3.2. Weaknesses

Although some participants found the array of transportation options available to users to be a strength, other participants found a lack of transit options to be a challenge for transportation. Interestingly, the Jeghers et al. [15] focus group findings indicated the cost of using rideshare services as a limitation and that community residents would like to experience more convenient travel, such as a light rail system to major cities throughout Florida. The lack of discussion regarding types of transit modes outside of driving a personal vehicle indicates that riders may benefit from education on optional mobility modes. The findings suggest that the transportation stakeholders demonstrate support and knowledge of these existing transportation technologies, such as autonomous rideshare or

micromobility options, but have not yet translated this to the users. As such, transportation stakeholders may work together to educate the community on the availability and operations of using various transportation modes to help increase community mobility and reduce disparities among mobility-vulnerable populations.

4.3.3. Opportunities

Discussion among the participants indicated that stakeholders would benefit from collaborating closely while integrating transportation technologies, such as autonomous ridesharing services. Collaborating with players in industry, academia, city and state officials, and community members is an important part of developing successful and sustainable changes for the transportation system [37]. By working together, stakeholders in the transportation system (i.e., designers, employees, and users) can enhance transportation in the community.

4.3.4. Threats

Participants discussed the critical threats to the transportation system, including the safety and regulatory policies for implementing innovative technologies and the impacts of public health emergencies (e.g., COVID-19). While implementing transportation technologies, such as autonomous rideshare services, cyber security and programming problems will continually concern users and operators [38]. Although implementing transportation technologies in the city may enhance the transportation system for people with mobility-vulnerable populations by increasing various modes and availability, the concerns of threats to security and health may be detrimental to this population.

Study participants presented SWOTs, such as the lack of services for vulnerable populations, safety, and transit impacts of COVID-19 that are like other Florida transit users [15]. Further, similar factors impacting the transportation system are also being experienced in other areas of the United States, such as funding availability [39,40]. As a result, identifying strengths, weaknesses, opportunities, and threats to the transportation system was comprehensive in that it aligned with research findings in other areas of the United States [41,42].

4.4. Limitations

Recruitment and communication with participants were limited to email, phone, and Zoom videoconferencing due to COVID-19 social-distancing guidelines, which may have impacted the rapport or dynamic between the team and participants. While literature suggests that data saturation can typically be achieved with 15 focus group participants, our study included only 13 participants, limiting the extent to which the full range of stakeholder perspectives and experiences were captured. The small sample size may impact the generalizability of findings and might not represent all community or stakeholder views. Additionally, responses could be influenced by participants' professional roles or organizational interests, introducing potential bias, and some participants may have withheld information due to concerns about their employers.

4.5. Strengths

Semi-structured interviews provided in-depth insights into stakeholders' perspectives on the transportation system, allowing for a comprehensive understanding of the topics. Engaging diverse stakeholders, including city officials, transportation providers, and community advocates, offered various viewpoints and contributed to a holistic view of the transportation system. Conducting interviews via Zoom facilitated interviews with stakeholders across the state.

4.6. Relevance

Study findings may encourage other transportation-related researchers to collaborate and identify transit needs, develop shared goals, and create educational resources that

assist mobility-vulnerable populations in accessing transportation. The study identifies the need for specialized funding and aligned priorities among transportation stakeholders. Policymakers may advocate for dedicated funding to support transit solutions for mobility-vulnerable populations, such as subsidized fares for low-income riders or expanding paratransit service hours to accommodate non-standard work schedules. The research findings indicate that resources exist for identifying user challenges and opportunities, yet further research is needed to elucidate why solutions are difficult to implement. Participants with advocacy backgrounds discussed resources and educational materials used to support mobility-vulnerable populations. For example, some participants discussed webinars and/or training sessions to teach older adults about alternative transportation options. Expanding these programs to include more tailored and frequent training sessions could improve accessibility. For example, developing easy-to-follow guides and user-friendly resources tailored to the needs of mobility-vulnerable people may help enhance confidence in using rideshares or public transit. Individuals with direct knowledge and experience with transportation in the community may advocate for solutions to issues impacting mobility-vulnerable populations. To comprehensively understand users' challenges and opportunities, consulting multiple stakeholders within the transportation system (e.g., city and state officials, industry employees, and healthcare workers) is essential. Future research may include a larger sample and a more diverse range of transportation stakeholders to further understand the broader implications of the challenges and opportunities identified in this study. Additionally, utilizing research methods such as a longitudinal approach [20] may help researchers assess the evolving impacts of the COVID-19 pandemic and emerging technologies on transportation systems for mobility-vulnerable populations.

One of the most critical takeaways from the study findings was the overlap of identified gaps, strengths, weaknesses, opportunities, and threats discussed among stakeholders. Regardless of whether the stakeholder was employed for Gainesville's city transportation system or represented mobility-vulnerable populations, they reported similar challenges (e.g., paratransit). While the research findings revealed some challenges and opportunities that may impact the translation of science, researchers may provide further insight and recommendations to address transportation-related disparities. This is particularly important to understanding transit needs and strategizing ways to mitigate transportation disparities among mobility-vulnerable populations.

5. Conclusions

The team captured valuable insight regarding transportation stakeholders' perceptions of the gaps, strategies utilized to meet the needs of transit users, and the transportation system's strengths, weaknesses, opportunities, and threats. This study uniquely captures challenges and opportunities from diverse transportation stakeholders, demonstrating shared and distinctive concerns that impact equitable transit access for mobility-vulnerable populations.

The main findings include the identification of transportation system gaps, such as the lack of funding and differing priorities across transportation stakeholders. The transportation gaps discussed impact transit services for mobility-vulnerable populations, such as the lack of reliable transportation options for everyday needs like healthcare, employment, education, and leisure. Findings indicate that advocacy for dedicated funding is needed to support mobility-vulnerable populations. Additionally, collaborative efforts among stakeholders, including city and state officials, industry representatives, and advocacy groups, may promote a more unified prioritization process aligning with short-term and long-term equitable transportation goals. Participants discussed strengths like willingness to pilot new technologies, weaknesses including unreliable transit options, opportunities for advancements with innovated technologies that may address disparities, and threats such as impacts of COVID-19 on transportation. Furthermore, stakeholders' willingness to pilot new technologies, such as autonomous ridesharing, presented an opportunity to miti-

gate service gaps. Future research may examine the efficacy of such emerging technologies in reducing disparities.

In this case study, the semi-structured interview demonstrates that transportation stakeholders are experiencing similar gaps, challenges, and opportunities to serve the greater community or specific population of focus. This study underscores the importance of collaborative action to address the evolving transportation needs of mobility-vulnerable populations. As such, researchers may work with other stakeholders (e.g., engineers and city and state officials) to share their knowledge of challenges and opportunities vulnerable populations experience to engage in everyday activities and promote quality of life.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/futuretransp4040073/s1>, Table S1: Semi-Structured Interview Guide.

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